Variable Subpixel Convolution Based Arbitrary-Resolution Hyperspectral Pansharpening

Lin He, *Member, IEEE*, Jinhua Xie, Jun Li, *Fellow, IEEE*, Antonio Plaza, *Fellow, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Jiawei Zhu

***Abstract*— Standard hyperspectral (HS) pansharpening relies on fusion to enhance low-resolution HS (LRHS) images to the resolution of their matching panchromatic (PAN) images, whose practical implementation is normally under a stipulation of scale invariance of the model across the training phase and the pansharpening phase. By contrast, arbitrary resolution HS (ARHS) pansharpening seeks to pansharpen LRHS images to any user-customized resolutions. For such a new HS pansharpening task, it is not feasible to train and store convolution neural network (CNN) models for all possible candidate scales, which implies that the single model acquired from the training phase should be capable of being generalized to yield HS images with any resolutions in the pansharpening phase. To address the challenge, a novel variable subpixel convolution (VSPC)-based CNN (VSPC-CNN) method following our arbitrary upsampling CNN (AU-CNN) framework is developed for ARHS pansharpening. The VSPC-CNN method comprises a two-stage elevating thread. The first stage is to improve the spatial resolution of the input HS image to that of the PAN image through a prepansharpening module, and then, a VSPC-encapsulated arbitrary scale attention upsampling (ASAU) module is cascaded for arbitrary resolution adjustment. After training with given scales, it can be generalized to pansharpen HS image to arbitrary scales under the spatial patterns invariance across the training and pansharpening phases. Experimental results from several specific VSPC-CNNs on both simulated and real HS datasets show the superiority of the proposed method.**

***Index Terms*— Arbitrary resolution, convolution neural network (CNN), hyperspectral (HS) pansharpening, spatial pattern, variable subpixel convolution (VSPC).**

Manuscript received 23 January 2022; revised 18 April 2022 and 31 May 2022; accepted 24 June 2022. Date of publication 11 July 2022; date of current version 29 July 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62071184, Grant 61571195, Grant 42030111, and Grant 61836003; in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022A1515011615; and in part by the Guangzhou Science and Technology Program under Grant 202002030395. *(Corresponding author: Jun Li.)*

Lin He, Jinhua Xie, and Jiawei Zhu are with the School of Automation Science and Engineering, South China University of Technology, Guangzhou 510640, China (e-mail: helin@scut.edu.cn).

Jun Li is with the School of Computer Science and the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: lijuncug@cug.edu.cn).

Antonio Plaza is with the Hyperspectral Computing Laboratory, University of Extremadura, 10003 Cáceres, Spain (e-mail: aplaza@unex.es).

Jocelyn Chanussot is with the Jean Kuntzmann Laboratory (LJK), National Centre for Scientific Research (CNRS), National Institute for Research in Digital Science and Technology (Inria), Grenoble Institute of Technology (Grenoble INP), Université Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@gipsa-lab.grenoble-inp.fr).

Digital Object Identifier 10.1109/TGRS.2022.3189624

NOMENCLATURE

$\mathbf{H}_t$	Low-resolution (observed) HS (LRHS) image for the training stage.
$\mathbf{H}_p$	LRHS image for the pansharpening stage.
$\mathbf{P}_t$	Observed PAN image for the training stage.
$\mathbf{P}_p$	Observed PAN image for the pansharpening stage.
$\mathbf{H}_t^{\downarrow}(\cdot)$	Reduced-resolution LRHS image for the training stage with a given scale.
$\mathbf{P}_t^{\downarrow}(\cdot)$	Reduced-resolution PAN image for the training stage with a given scale.
$\widehat{\mathbf{H}}_s$	Reconstructed HS image for standard pansharpening (with the same spatial resolution as $\mathbf{P}_p$).
$\widehat{\mathbf{H}}_{s_a}$	Reconstructed ARHS image.
s	Pansharpening scale of standard pansharpening (i.e., spatial resolution ratio between observed PAN and LRHS).
s_a	ARHS pansharpening scale (i.e., the spatial ratio between ARHS pansharpened image and the LRHS image $\mathbf{H}_p$).

I. INTRODUCTION

HYPERSPECTRAL (HS) images, which capture hundreds of narrow spectral bands in wide spectral coverage, have been used for various applications, such as classification [1]–[3], change detection [4], object detection [5], and anomaly detection [6]. However, the spatial resolutions of HS images are usually limited owing to the physical constraints of optical systems, which restricts the scope of their practical application, while imaging sensors can easily acquire high spatial resolution panchromatic (PAN) images over the spectral range of the visible light. With the matching PAN images being available, a feasible way to enhance the resolutions of HS images is to resort to HS pansharpening [7], which refers to fusing low-resolution HS (LRHS) image and high-resolution (HR) PAN image to generate an enhanced image with both high spatial and high spectral resolutions.

HS pansharpening, which is extended from traditional multispectral (MS) pansharpening [8]–[10], has started to attract attention over recent years. Some of the methods initially designed for MS pansharpening can be used as universal ones to deal with HS pansharpening. For example, several

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

methods based on CS and MRA were utilized for sharpening HS images [11]. CS-based methods involve domain transformations to separate spatial components of the original LRHS image and accordingly make replacement [12], [13]. MRA-based methods extract spatial detail of the PAN image via multiresolution decompositions and then inject it into HS data [14]–[16]. In addition, Bayesian methods and matrix factorization methods have been developed for HS pansharpening. Bayesian methods formulate the pansharpening process as an optimization problem in the Bayesian inference framework and regularize it by defining an appropriate prior distribution [17], [18]. Matrix factorization methods are based on linear mixture models, which alternately unmixes HS and PAN to obtain the endmember spectra and abundance map [19]. Recently, convolution neural networks (CNNs) for the pansharpening task have been explored and achieved state-of-the-art performance, which often trains models at the reduced resolution (RR) and then, under the hypothesis of scale invariance of the CNN model, applies them to the full resolution (FR) to generate the final pansharpening result [9], [20]. Masi *et al.*. pioneerly established a pansharpening CNN with a three-layer structure [20]. Wei *et al.*. [21] employed the residual learning to establish a deeper CNN (DRPNN) for pansharpening. By carefully considering spectral preservation and inherent structures such as detail injection, He *et al*. [22], [23] designed spectral fidelity oriented HS pansharpening CNNs. Zheng *et al.*. [24] built a CNN method for HS pansharpening that preinterpolated the LRHS images through an HS prior and designed channel and spatial attention into residual blocks for adaptively feature learning. Dong *et al.*. [25] applied the Laplace pyramid technique to decompose PAN image details into different levels, and each level was processed by a spatial dense network for extracting details, which were then fused with the upsampled HS image. Qu *et al*. [26] designed a multiresolution residual network to extract spatial–spectral features for reconstructing HS images.

Over past two decades, pansharpening has been used to assist some practical remote sensing tasks, including thematic mapping enhancement [27]–[29], agriculture management [30]–[32], object detection [33], [34], and visual interpretation [7], [35]. For these tasks, pansharpening is usually involved as a preprocessing phase where data resolution is regulated for the final applications. However, the required spatial resolutions in some real-world applications, such as the aforementioned, may not be the same as that of the PAN image. Standard HS pansharpening, which aims at promoting the spatial resolution of an HS image to that of the matching PAN, is obviously incapable of dealing with such a situation. In contrast, arbitrary resolution HS (ARHS) pansharpening, which seeks to adjust the HS resolution to any desired one, poses a feasible way to meet the flexible requirements of possible practical applications with any user-customized resolutions.

For ARHS pansharpening, one of the most challenging obstacles to handle is how to generate HS images with arbitrary resolutions or unpredefined ones using a single trained model. Within the context of practical implementations, standard learning-based HS pansharpening (including normal CNN-based) is usually composed of two major stages. The first one is the RR training phase. Therein, the observed HS and the connected PAN images go through a degradation process to generate RR ones, on which a pansharpening CNN model is trained with the supervising of the observed HS image, which is used as the reference image. The other stage is a prediction or pansharpening procedure to yield the final FR pansharpening result. Such a pansharpening phase, under the stipulation of scale invariance of the CNN model, utilizes the pansharpening model trained in the preceding RR stage to integrate the upcoming LRHS and PAN images for generating the FR HS images, which are desired pansharpened ones. It is worth noting that, in such a standard pansharpening task, the pansharpening scale (i.e., the resolution ratio of before and after pansharpening HS data) is fixed across the training and the pansharpening stages. However, for ARHS pansharpening, there may be many possible desired pansharpening resolutions, which means that the CNN model trained under one or several given scales has to be used to deal with unexpected upcoming scale requirements in the pansharpening phase. Therefore, how to pansharpen HS images to arbitrary resolutions (including integer and noninteger scales) beyond the training scales using a single pansharpening model is one of the main challenges for ARHS pansharpening, which does not arise in standard HS pansharpening.

In this work, we propose a variable subpixel convolution (VSPC)-based CNN (VSPC-CNN) method for the ARHS pansharpening task by following our arbitrary upsampling CNN (AU-CNN) framework. Our method is able to generalize a CNN model trained under certain scales to pansharpen HS data for any scales, thus overcoming the dilemma of how to reconstruct HS images with arbitrary resolutions using just a single CNN model. More specifically, the innovative contributions of our work can be identified as follows.

1) *AU-CNN Framework:* The framework is based on the AU strategy and is in fact a general pansharpening scheme, which can be used to guide the design of new ARHS pansharpening CNNs. It is a two-stage elevating process led by a relatively strict mathematical representation. The first stage is associated with a prepansharpening subnetwork to elevate the spatial resolution of input data to that of the PAN image, while, in the second stage, an AU-involved subnetwork is cascaded for a further resolution elevating.
2) *Innovative VSPC for Learnable AU:* In order to accomplish upsampling with an arbitrary scale including integer and noninteger, we design the VSPC operation that inserts a variable number of zeros in variable positions of low-resolution data to create a subpixel image of arbitrary resolution and then fulfill convolution.
3) *VSPC-CNN ARHS Pansharpening Method:* This CNN method is designed by following our AU-CNN framework. It is composed of a prepansharpening module that can be set as standard HS pansharpening CNN and an arbitrary scale attention upsampling (ASAU) module, which encapsulates the VSPC unit to flexibly adjust the data resolution. Specifically, the ASAU module

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

encapsulates VSPC to perform AU and integrates a scale spatial attention (SSA) block that introduces the scale prior to capturing more spatial information. For our VSPC-CNN method, there exists spatial patterns invariance across pansharpening scales. This ensures the generalization ability of the CNN model across the training and the pansharpening phase. Therefore, after trained with fixed scales, the CNN model can be directly used to pansharpen HS images to arbitrary resolutions.

The rest of this article is organized as follows. The proposed AU-CNN framework, the VSPC operation, and the VSPC-based ARHS pansharpening CNN are presented in detail in Section II. In Section III, the connected experimental results are given and analyzed. Finally, the conclusion is drawn in Section IV.

II. Proposed Method

In this section, we develop a VSPC-based ARHS pansharpening CNN following an AU-CNN framework. In addition, the associated mechanism of scale generalization is further justified.

ARHS pansharpening aims to utilize an LRHS image $\mathbf{H} \in \mathbb{R}^{m\times n\times b}$ and the connected PAN image $\mathbf{P} \in \mathbb{R}^{sm\times sn}$ to achieve an arbitrary resolution pansharpened HS image $\widehat{\mathbf{H}}_{s_a} \in \mathbb{R}^{s_a m\times s_a n\times b}$, where $m\times n$ represents the spatial size of the LRHS image, b denotes the number of its spectral channels, s denotes the spatial resolution ratio between the PAN and the LRHS images, which is, in fact, the prepansharpening scale in our method, s_a is the ARHS pansharpening scale being the spatial resolution ratio between the ARHS and the LRHS images, and $s' = s_a/s$ is an intermediate scale being the spatial resolution ratio between the ARHS and the PAN images. s_a can be any value determined by the practical arbitrary resolution requirement, either integer or noninteger, usually being greater than s. When s_a equals s, ARHS pansharpening is degenerated to standard HS pansharpening.

For convenient use later on, Nomenclature illustrates the main symbols and their brief descriptions used in this article.

A. Overall Design

CNN-based standard HS pansharpening seeks to integrate LRHS and PAN images to produce HR HS images with the spatial resolution of PANs by using a CNN structure. Its practical fulfillment is normally composed of a training phase and a pansharpening phase, the former of which can be performed on RR data. The training and pansharpening phases can be formulated as

$$\boldsymbol{\theta}_{\text{opt}} = \arg\min_{\theta} \mathcal{L}\left(\Phi\left(\mathbf{H}_t^{\downarrow}(s), \mathbf{P}_t^{\downarrow}(s); \boldsymbol{\theta}\right), \mathbf{H}_t\right) \tag{1}$$

and

$$\widehat{\mathbf{H}}_s = \Phi\left(\mathbf{H}_p, \mathbf{P}_p; \boldsymbol{\theta}_{\text{opt}}, s\right) \tag{2}$$

respectively, where $\mathcal{L}(\cdot)$ is the pansharpening loss, $\Phi(\cdot)$ represents the CNN mapping between the input HS image and the target image, θ denotes the CNN parameters to be optimized (i.e., the CNN model), and $(r(\mathbf{H}_t)/r(\mathbf{H}_t^{\downarrow}(s))) = (r(\mathbf{P}_t^{\downarrow}(s))/r(\mathbf{H}_t^{\downarrow}(s))) = (r(\mathbf{P}_t)/r(\mathbf{H}_t)) = s$ if we let $r(\cdot)$ indicate the spatial resolution of an image. Once the CNN parameters $\boldsymbol{\theta}$ are determined as $\boldsymbol{\theta}_{\text{opt}}$ after training, the model $\boldsymbol{\theta}_{\text{opt}}$ can be used to produce the pansharpened HS data for upcoming LRHS and PAN images. In contrast to standard HS pansharpening, ARHS pansharpening, as an innovative pansharpening task, aims to generate arbitrary scale pansharpened HS image, which can mathmatically represent its training and pansharpening phases as

$$\boldsymbol{\theta}_{\text{opt}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}\left(\Phi\left(\mathbf{H}_t^{\downarrow}(S), \mathbf{P}_t^{\downarrow}(S); \boldsymbol{\theta}\right), \mathbf{H}_t\right) \tag{3}$$

and

$$\widehat{\mathbf{H}}_{s_a} = \Phi\left(\mathbf{H}_p, \mathbf{P}_p; \boldsymbol{\theta}_{\text{opt}}, s_a\right) \tag{4}$$

respectively, where S denotes training scales and s_a represents the target scale, which can be arbitrarily set.

To deal with such a new type of pansharpening task, an ARHS pansharpening method needs to be carefully designed. An intuitive way to arbitrarily adjust the spatial resolution of data is to rely on the AU operation, which could be either interpolation-based or learning-based. Meanwhile, the PAN image collaborating with the LRHS image should be integrated in a process of enhancing the LRHS data to the spatial resolution of its own, considering full and direct utilization of the PAN's spatial structures. Therefore, the above CNN-based ARHS pansharpening mentioned in (3) and (4) can be reformulated as a two-stage elevating form

$$\boldsymbol{\theta}_{\text{opt}}, \boldsymbol{\theta'}_{\text{opt}} = \arg\min_{\boldsymbol{\theta},\boldsymbol{\theta'}} \mathcal{L}\left(\mathcal{AU}\left(\Psi\left(\mathbf{H}_t^{\downarrow}(S), \mathbf{P}_t^{\downarrow}(S); \boldsymbol{\theta}\right), \boldsymbol{\theta'}\right), \mathbf{H}_t\right) \tag{5}$$

and

$$\widehat{\mathbf{H}}_{s_a} = \mathcal{AU}\left(\Psi\left(\mathbf{H}_p, \mathbf{P}_p; \boldsymbol{\theta}_{\text{opt}}\right), \boldsymbol{\theta'}_{\text{opt}}, s_a\right) \tag{6}$$

respectively, where $\Psi(\cdot)$ is a prepansharpening module that is used to enhance the spatial resolution of the input LRHS to that of the PAN for fully exploiting the spatial information of the PAN and $\mathcal{AU}(\cdot)$ indicates an AU operation that can be interpolation-based or learning-based. Therefore, we have an AU-CNN framework, as formulated in (5) and (6), and graphically illustrated in Fig. 1, which is able to guide the design of novel ARHS pansharpening CNNs. In the training phase of our AU-CNN framework, training data of limited number of scales S (i.e., a scale set) are used. At each iteration, one scale was randomly sampled from S for training. After training, we obtain a pansharpened CNN model trained on some scales. Then, the trained CNN model can be used to pansharpen the HS image to any target or desired resolutions. It is noteworthy that the possible target resolutions or scales may be far beyond the scales in the scale set S used for training. Therefore, in Fig. 1, the one stack shown in the pansharpening or testing phase refers to one of any possible pansharpening scales, while multiple stacks shown in the training phase mean that one or more scales of data could be included in this phase. The prepansharpening module can be set as a standard HS pansharpening CNN, which, on the one hand, meets the requirement of yielding intermediate data with the PAN resolution and, on the other hand, is beneficial to transplant

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

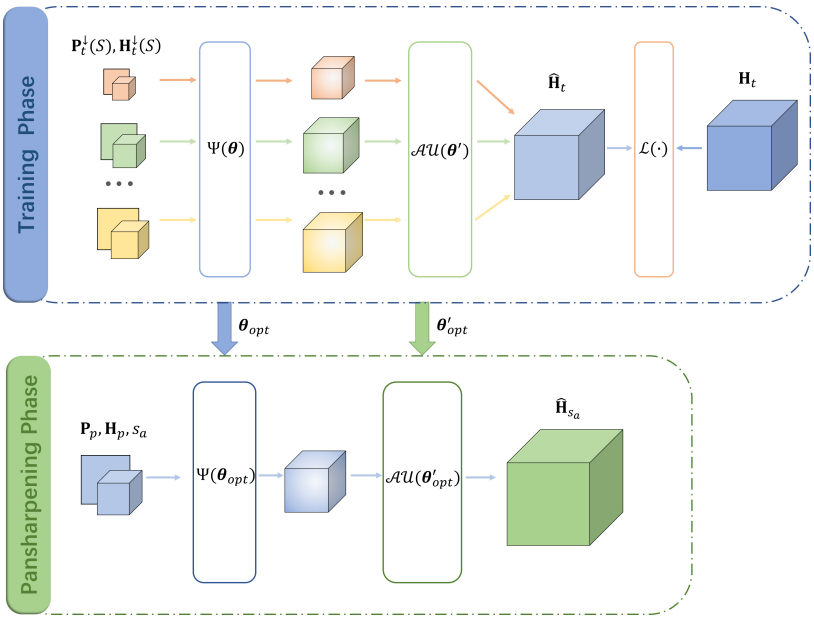


Fig. 1. Block diagram of our AU-CNN framework.

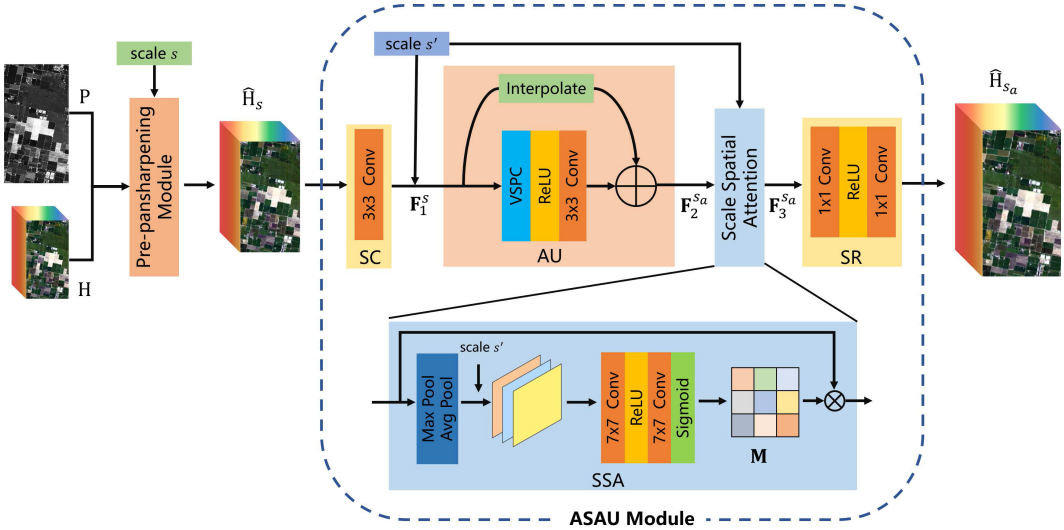


Fig. 2. Overall architecture of our VSPC-based ARHS pansharpening CNN.

existing pansharpening CNNs. In our AU-CNN framework, subsequently connected to the prepansharpening module is the AU operation. For the upsampling operation $\mathcal{AU}(\cdot)$, traditional image interpolation methods, such as nearest-neighbor interpolation, bilinear, and bicubic interpolation, can be directly used. However, those traditional interpolation-based upsampling methods improve the image resolution with fixed kernels that are not trainable, which often introduces some side effects, such as noise amplification and blurry results. Alternatively, in order to overcome the weaknesses of interpolation-based methods and learn upsampling in an end-to-end manner, transpose convolution [36], [37] or subpixel convolution [38] can be applied, which have been utilized in some image upsampling tasks, such as image super-resolution. Unfortunately, they can only work for integer scales, which does not meet the requirement of arbitrary-resolution adjustment in our ARHS pansharpening task.

Therefore, following the AU-CNN framework constructed above, in this work, we develop a novel VSPC that can implement AU and further a VSPC-CNN for the ARHS pansharpening task. This CNN is, thus, mainly composed of a prepansharpening module and an ASAU module that encapsulates VSPC. Fig. 2 graphically illustrates the overall architecture of the proposed VSPC-based ARHS pansharpening CNN.

B. Variable Subpixel Convolution

The subpixel convolution operation, which fulfills convolution at a subpixelwise scale, has been successfully applied in traditional image enhancement tasks, as a way to perform learnable upsampling with integer scales [38]–[42]. Usually, there are two paths for its implementation. The first one is efficient subpixel convolution (ESPC) where multiple normal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

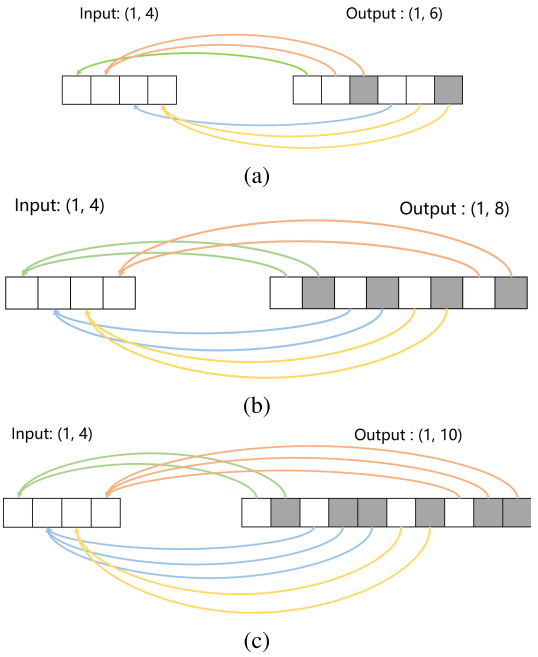


Fig. 3. Subpixel space formed from the LR input, where gray pixels denote those set to zero. (a) Upsampling scale $r = 1.5$. (b) Upsampling scale $r = 2.0$. (c) Upsampling scale $r = 2.5$.

convolution operations are carried out simultaneously and then follow by pixel shuffling. The other one is direct subpixel convolution (DSPC), which views subpixel convolution as an operation in a subpixel space and, therefore, establishes a subpixel image followed by a normal convolution. However, both ESPC and DSPC can only fulfill integral upsampling for image super-resolution, i.e., enhance image resolution with integer ratio, owing to the way of uniformly zero inserting in DSPC and generating integral multiple channels in ESPC. This means that they are unable to tackle the noninteger scales, which may arise in ARHS pansharpening. In addition, subpixel convolution-based image super-resolution methods, such as ESPC and DSPC, work under the stipulation of scale invariance of the CNN model, which is not designed to solve the scale generalization in ARHS pansharpening task (i.e., how to generate HS images with arbitrary resolution beyond the training scale). Therefore, traditional subpixel convolution methods, such as ESPC and DSPC, cannot be applied for the ARHS pansharpening task.

In this work, we design a VSPC to perform arbitrary resolution enhancement for the ARHS pansharpening task. It creates subpixel images by inserting a variable number of zeros in variable positions according to coordinate transforms under required zoom ratios or upsampling scales. A subpixel image is created from the original LR image, and hence, their coordinates are related to each other. For each pixel (i, j) in the subpixel space, we use the following coordinate transform operator to map it back to the LR space:

$$\mathcal{C}((i, j), s) = \left(\left\lfloor \frac{i + 0.5}{s} \right\rfloor, \left\lfloor \frac{j + 0.5}{s} \right\rfloor \right) \tag{7}$$

where $\mathcal{C}(\cdot, \cdot)$ represents the coordinate transform between the subpixel space coordinate and the LR space coordinate, s is the upsampling scale (which can be integer or noninteger), and $\lfloor \cdot \rfloor$ is the floor function. Note that 0.5 is added to align the geometric center of the LR image and that of the subpixel image. Let F_{LR} denote the feature maps that need to be upsampled, and F_{HR} is high spatial resolution feature maps. Then, our VSPC can be formulated as follows:

$$\begin{aligned} F_{\mathrm{HR}} &= W^{\mathrm{SP}} * \mathcal{CSP}(F_{\mathrm{LR}}, s) \\ &= W^{\mathrm{SP}} * (\mathcal{ZI}(\mathcal{C}(F_{\mathrm{LR}}, s))) \end{aligned} \tag{8}$$

where $\mathcal{CSP}$ means creating subpixel space from LR feature maps, $\mathcal{ZI}(\cdot)$ represents the zero-inserting operation, and W^{SP} denotes convolution kernels used in subpixel space.

Fig. 3 graphically exemplifies three 1-D upsampling cases under different upsampling scales for our VSPC. If the scale s is set to an integer 2, then, after the coordinate transform, every two pixels in the subpixel space are mapped to the same positions in the LR space. However, if the scale s is set to a noninteger, such as 1.5 and 2.5, the transform results are more complicated. The number of pixels in the subpixel space that is mapped to the same LR space pixel ranges from $[1, \mathrm{ceil}(s)]$. If the number is n, which is more than 1, it indicates that the pixel in LR space is related to n pixels in subpixel space. Accordingly, we should insert $n - 1$ (rows and columns) zeros (subpixels) into the neighborhood of that LR space pixel, which is variable in quantity and position. That is, VSPC is able to create a subpixel image of arbitrary resolution given the upsampling scale s, no matter whether it is an integer or a noninteger, and then to perform convolution on the subpixel image to accomplish AU.

In Section II-D, we will justify the ability of our ARHS pansharpening method encapsulating VSPC to generalize a single model trained under certain scales to any pansharpening scale.

C. Arbitrary Scale Attention Upsampling Module

In our AU-CNN framework mentioned in Section II-A, how to integrate our proposed VSPC needs to be carefully designed. On the one hand, directly putting VSPC into the prepansharpening module may break original network structures, such as global residual connections. On the other hand, VSPC may produce some all-zero regions as the pansharpening scale increases, which would negatively affect the pansharpening performance. Besides, it is noteworthy that the upsampling scale can be used as useful prior information to promote the ARHS pansharpening performance, whereas, in VSPC, the upsampling scale is only used for creating subpixel images. Finally, directly performing convolution after prepansharpening involves many parameters, tending to incur a high computation load. Therefore, we design an ASAU module encapsulating VSPC to perform AU, instead of directly using VSPC. The module first compresses the spectral dimension and then integrates VSPC into a residual block for all-zero regions' compensation. In addition, an SSA block involving the consideration of prior upsampling scales is constructed for adaptively adjusting the importance of various regions. Finally, a spectral reconstruction (SR) block is used to recover data with the original spectral dimension. The architecture of our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

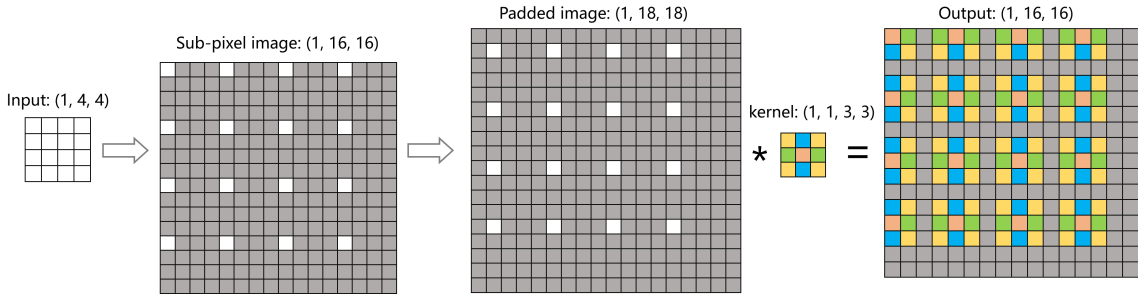


Fig. 4. Proposed VSPC when upsampling scale $s' = 4$ and convolutional kernel size $k = 3$, where gray pixels denote those set to zero.

ASAU module is shown in Fig. 2. Blocks of the module are described in detail as follows.

1) *Spectral Compression (SC):* HS images have the characteristic of high correlation among neighboring bands [43]. Performing reduction of spectral dimension before upsampling would decrease computation load and parameters of subsequent processing. We use d 3×3 convolutions to reduce the spectral dimension down to d ($d < b$), which can be formulated as follows:

$$\mathbf{F}_1^s = f_3\left(\widehat{\mathbf{H}}_s\right) \tag{9}$$

where $\widehat{\mathbf{H}}_s \in \mathbb{R}^{sm\times sn\times b}$ denotes the prepansharpening result, $f_3(\cdot)$ indicates the 3×3 convolution for SC, and $\mathbf{F}_1^s \in \mathbb{R}^{sm\times sn\times d}$ is dimension-reduced features with the PAN spatial resolution.

2) *AU:* After SC, our proposed VSPC can be applied for AU. Since we aim at implementing AU, the upsampling scale s' may be larger than the convolutional kernel size k. Specifically, when the number of zeros that VSPC inserts to create a subpixel image is equal or greater than k, some all-zero regions in the subpixel image will be equal or greater than the kernel size. In this case, the convolution on the subpixel image could not learn useful features from those regions and result in some all-zero rows and columns. The situation is shown in Fig. 4. Therefore, we need to make compensation for these all-zero rows and columns. An intuitive way is to add a shortcut connection between the input and output of VSPC. Inspired by the residual block designed by He *et al.* [44], we design a residual-block-like VSPC block named AU, as shown in Fig. 2. In AU, the shortcut connection is linked from its bottom and goes through a bicubic interpolation before getting to its top. Thus, the output of AU block can be formulated as

$$\mathbf{F}_2^{s_a} = f_k\left(\delta\left(W_k^{\mathrm{SP}} * \mathcal{CSP}\left(\mathbf{F}_1^s, s'\right)\right)\right) + \uparrow \mathbf{F}_1^s \tag{10}$$

where $\mathcal{CSP}(\cdot, s')$ represents creating subpixel image for arbitrary scale s', W_k^{SP} is the convolution of kernel size k, $f_k(\cdot)$ is the second $k \times k$ convolution layers, $\delta(\cdot)$ represents the ReLU function, $\uparrow$ indicates the upsampling interpolation, and $\mathbf{F}_2^{s_a} \in \mathbb{R}^{s_a m\times s_a n\times d}$ denotes the feature maps with the target spatial resolution (note that $s_a = s \times s'$).

3) *SSA:* For ARHS pansharpening, different scales are related to different downsampling degradations, which implies that including the scale factor in CNN learning has a beneficial effect on our pansharpening. Hence, we design an SSA block, as shown in Fig. 5, to generate spatial attention maps adaptively modulating informative regions connected to upsampling scales. More specifically, the SSA map $\mathbf{M}$ is first computed as

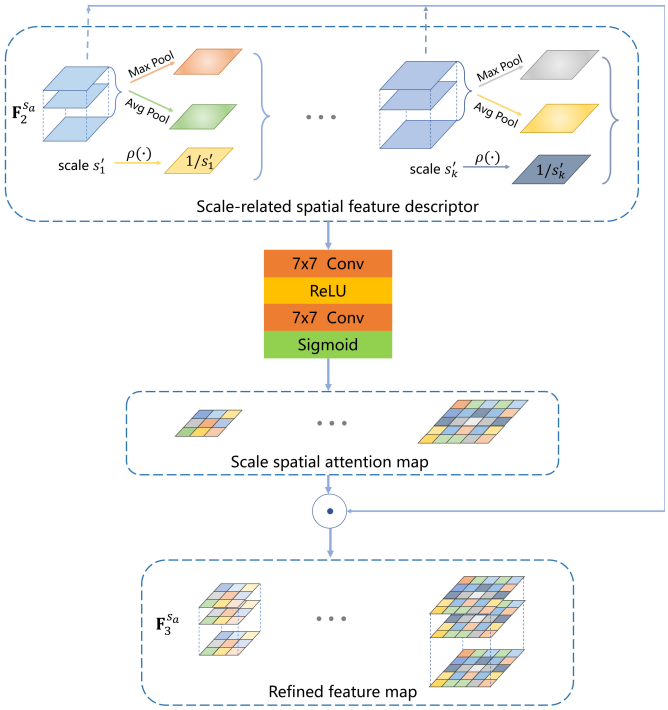


Fig. 5. Detailed architecture of the SSA block.

$$\mathbf{M} = \sigma f_7\left(f_7\left(\left[\mathrm{AvgPool}\left(\mathbf{F}_2^{s_a}\right); \mathrm{MaxPool}\left(\mathbf{F}_2^{s_a}\right); \rho\left(s'\right)\right]\right)\right) \tag{11}$$

where AvgPool$(\cdot)$ and MaxPool$(\cdot)$ denote the global average-pooling and max-pooling along the channel axis, respectively, and $\rho(s')$ indicates generating a 2-D map of all $1/s'$, which has the same spatial size as $\mathbf{F}_2^{s_a}$. That is, 2-D maps yielded by the average-pooling, the max-pooling, and the scale operation $\rho(s')$ are concatenated to generate scale-related spatial feature descriptors, and then, spatial convolutions $f_7(\cdot)$ and the sigmoid function $\sigma(\cdot)$ are collaborated to generate SSA maps. Based on the SSA maps, the refined feature map

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

$\mathbf{F}_3^{s_a} \in \mathbb{R}^{s_a m \times s_a n \times d}$ is computed as

$$\mathbf{F}_3^{s_a} = \mathbf{M} \odot \mathbf{F}_2^{s_a} \tag{12}$$

where $\odot$ denotes the elementwise multiplication.

4) *SR:* After AU and scale attention refinement, we need to perform an SR on the intermediate feature maps to produce a data cube with the same spectral bands as the original HS image. As shown in Fig. 2, our SR module consists of two 1×1 convolutions and an activation function ReLU. 1×1 convolutions are used here to exclusively adjust spectral dimension while preserving the spatial information. Thus, the reconstructed ARHS image $\widehat{\mathbf{H}}_{s_a} \in \mathbb{R}^{s_a m \times s_a n \times b}$ can, finally, be acquired as

$$\widehat{\mathbf{H}}_{s_a} = f_1\left(\delta\left(f_1\left(\mathbf{F}_3^{s_a}\right)\right)\right) \tag{13}$$

where $f_1(\cdot)$ denotes the 1×1 convolution.

It is noteworthy that, when $s_a = s$, ARHS pansharpening has degenerated to standard HS pansharpening, and thus, the resolution of the input of ASAU is the same as that of its output. In this situation, the coordinate transform of our VSPC integrated into the ASAU module becomes the identity transform, which implies that the zero-inserting operation is not required. This means that the subpixel convolution operation in VSPC has degenerated to a normal convolution, and therefore, ASAU is degenerated to a block just offering some additional normal convolution operations for standard pansharpening.

Algorithm 1 VSPC-Based ARHS Pansharpening CNN

Input:

HS image and its connected PAN image for training phase $\mathbf{H}_t$, $\mathbf{P}_t$;
The set of selected training scales S;
HS image and its connected PAN image for pansharpening phase $\mathbf{H}_p$, $\mathbf{P}_p$;
The user-customized scale s_a

Output:

HS image on the user-customized scale $\widehat{\mathbf{H}}_{s_a}$;

Synthesize the reduced-resolution HS images and the corresponding PAN images $\mathbf{H}_t^{\downarrow}(S), \mathbf{P}_t^{\downarrow}(S)$ by degrading $\mathbf{H}_t$, $\mathbf{P}_t$ according to training scales S;
Initialize model parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ randomly;
for epoch in N epochs **do**
  Sample LR-HR batch for scale $s_{ai} \in S$;
  Obtain $\widehat{\mathbf{H}}_s$ by pre-pansharpening module;
  Obtain $\mathbf{F}_1^s$ by SC block;
  Create sub-pixel image with a certain group of spatial patterns according to scale s_{ai};
  Obtain upsampling feature maps $\mathbf{F}_2^{s_a}$ for scale s_{ai} using Eq. 10;
  Obtain refined feature maps $\mathbf{F}_3^{s_a}$ by SSA block;
  Obtain $\widehat{\mathbf{H}}_{s_a}$ by SR block;
  Calculate the loss function $\mathcal{L}$;
  Update parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ with respect to $\mathcal{L}$;
end for
Obtain optimized model parameters $\boldsymbol{\theta}_{opt}$ and $\boldsymbol{\theta}'_{opt}$;
return $\widehat{\mathbf{H}}_{s_a} = \mathcal{AU}(\Psi(\mathbf{H}_p, \mathbf{P}_p; \boldsymbol{\theta}_{opt}), \boldsymbol{\theta}'_{opt}, s_a)$;

D. Spatial Pattern Invariance Across Pansharpening Scales

Standard HS pansharpening treats pansharpening of different integer scales as independent tasks, and therefore, it needs to train a specific model for each scale. However, the target resolution of ARHS pansharpening is arbitrary (often user-customized), where the number of possible desired scales is much more than that of standard pansharpening. If training a specific model for each scale, it will result in expensive computation and heavy memory burden, even impossible fulfillment. Therefore, it is crucial for HS pansharpening to pansharpen upcoming HS data of arbitrary scale using a single CNN model obtained with training data of limited scales. Our proposed VSPC-CNN is capable of handling the challenge due to its generalization ability. In the following, we will justify the ability.

In order to reconstruct HS images to any spatial resolutions, our VSPC-CNN first creates subpixel images of training scales, then trains a CNN model on them, and, finally, uses the trained model to fulfill pansharpening with any scales. As illustrated in Fig. 6, zero-inserting operation forms certain spatial patterns for convolution. Under a fixed mask size, the number of spatial patterns is limited, no matter how the pansharpening scale varies. That is, there exists spatial patterns invariance across pansharpening scales in our VSPC-CNN. During the training phase of this CNN, these spatial patterns are involved in learning the connected CNN model. When we, in the pansharpening phase, use this CNN model to pansharpen upcoming data to arbitrary resolutions, the spatial patterns still work, i.e., these spatial patterns keep consistent across the training and pansharpening phases of ARHS pansharpening. Fig. 7 illustrates the flowchart of training and pansharpening phases for our VSPC-CNN. Therefore, the CNN model obtained with a certain group of spatial patterns in the training phase can be generalized to the pansharpening phase, where ARHS images are yielded using convolutions on the associated subpixel images, which always comprises the same group of spatial patterns.

Our VSPC-based ARHS pansharpening CNN designed with the considerations in this section offers a way to deal with the new ARHS pansharpening task using the upsampling strategy. Its practical implementation is summarized in Algorithm 1.

III. Experimental Results and Analysis

A. Experimental Dataset

In this section, the proposed method is tested on three HS datasets: Pavia Center, Chikusei, and Merced, whose descriptions are listed in Table I. Pavia Center and Chikusei are used as simulated datasets, while Merced is used as a real dataset. Simulated datasets refer to those datasets that only have observed HS images but without connected PAN images. In this case, the PAN images need to be simulated from the observed HS images. In contrast to simulated datasets, real datasets comprise both the observed HS image and connected real PAN image, and thus, simulated processes are not required.

1) *Pavia Center Dataset:* The Pavia Center contains HS data with spatial size 1096*715, which was

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

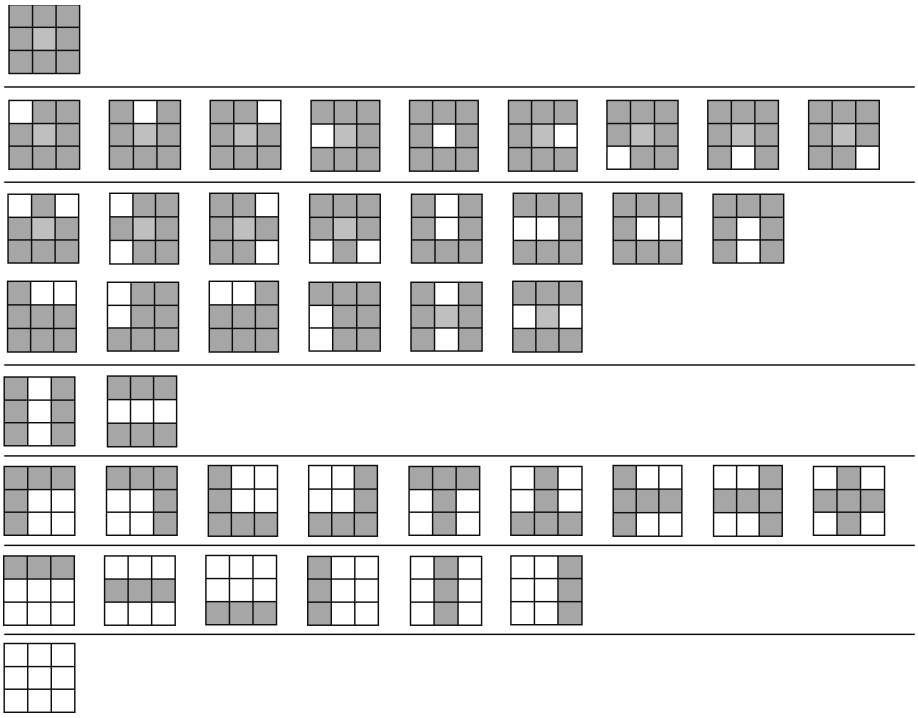


Fig. 6. Convolutional spatial patterns when upsampling scales r' range from [1.0, 2.5] with interval 0.1 and convolutional kernel size $k = 3$. Gray pixels denote inserted zeros.

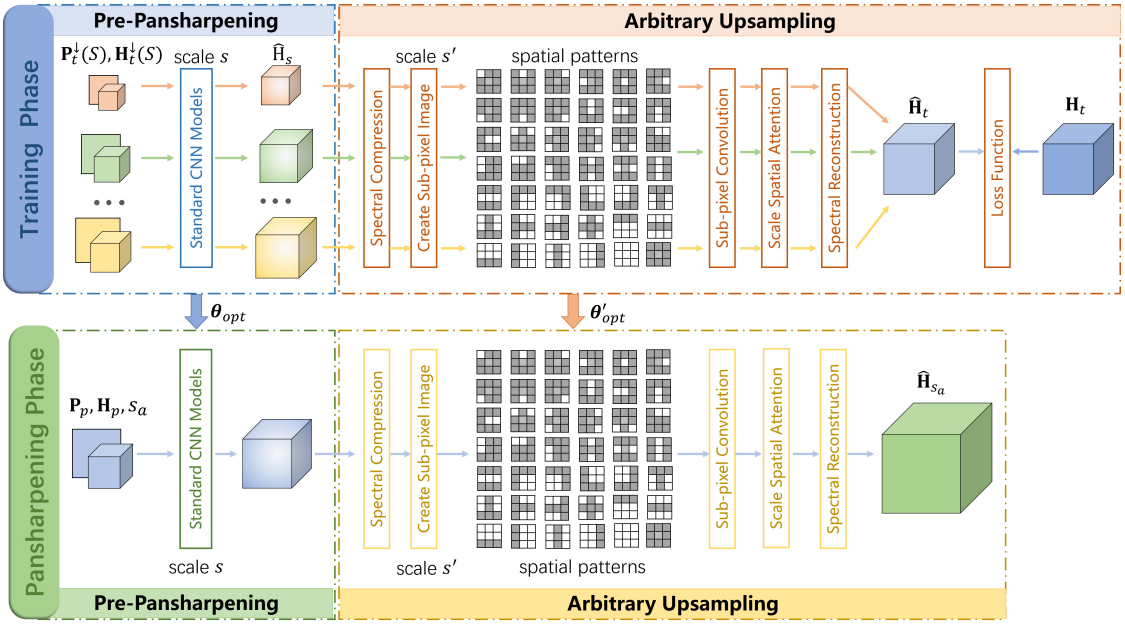


Fig. 7. Flowchart of training and pansharpening phases for our VSPC-CNN.

TABLE I
PROPERTIES OF THE DATASETS USED FOR EVALUATION

Datasets	Spectral Range	Train HS	Train PAN	Test HS	Test PAN	Instrument
Pavia Center	0.4-0.8 μm	HS 200*40*102	PAN 1000*200	HS 40*40*102	PAN 200*200	ROSIS
Chikusei	0.4-1.0 μm	HS 100*100*124	PAN 500*500	HS 40*40*124	PAN 200*200	Headwall Hyperspec-VNIR-C
Merced	0.4-2.5 μm	HS 200*60*134	PAN 600*180	HS 60*60*134	PAN 180*180	Hyperion and ALI

acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor during a flight campaign over Pavia, Northern Italy. It contains 115 spectral bands that range from the wavelengths of 430 to 860 nm, and 13 water absorption bands have been discarded.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

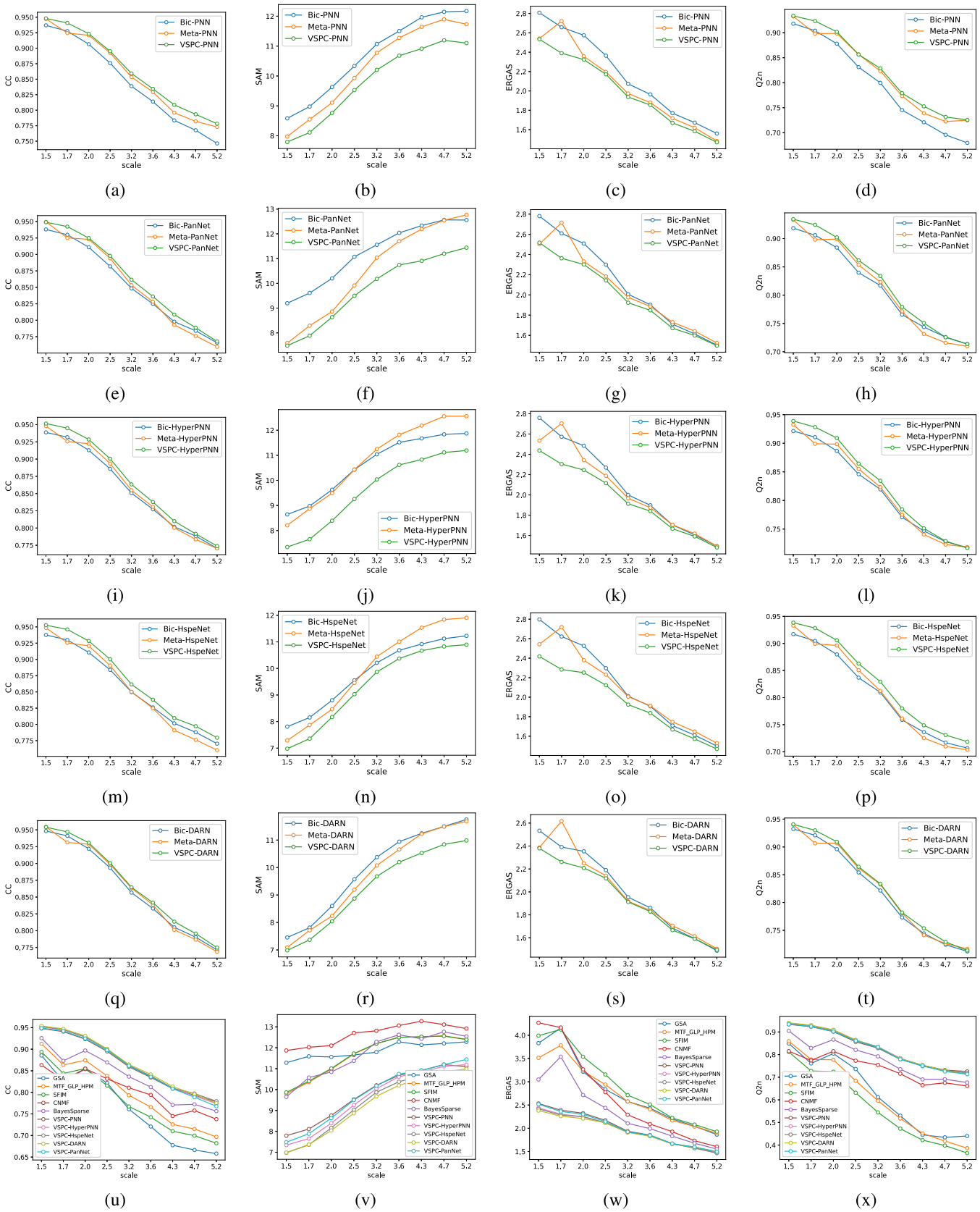


Fig. 8. Pansharpening performances under arbitrary scale s_a on the Pavia Center dataset: (a)–(d), (e)–(h), (i)–(l), (m)–(p), and (q)–(t) are the comparative results of five specific VSPC-CNNs, respectively; (u)–(x) represent the results of all compared methods.

2) *Chikusei Dataset:* The Chikusei contains HS data with spatial size 2517*2335, which was taken by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

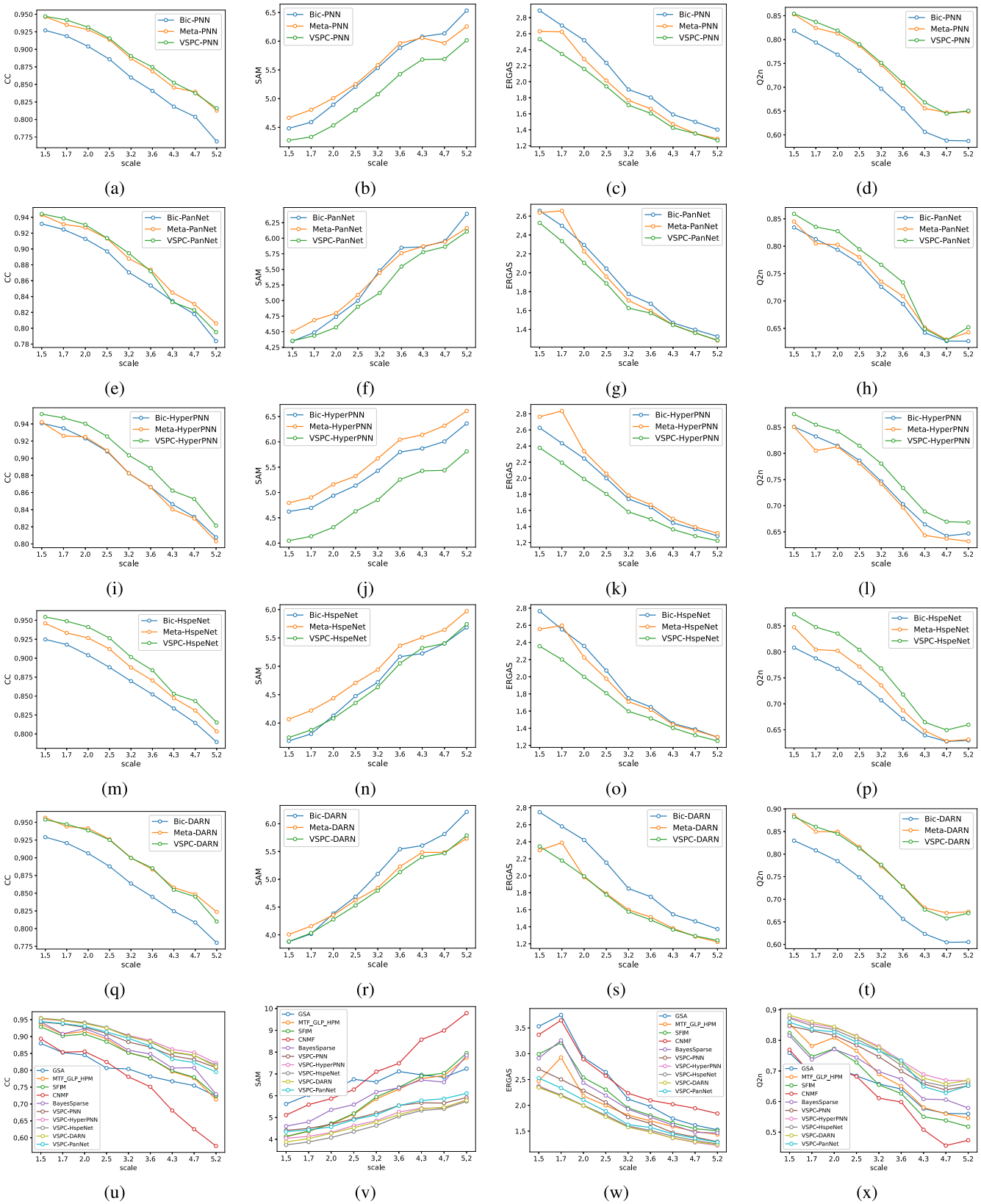


Fig. 9. Pansharpening performances under arbitrary s_a on the Chikusei dataset: (a)–(d), (e)–(h), (i)–(l), (m)–(p), and (q)–(t) are the comparative results of five specific VSPC-CNNs, respectively; (u)–(x) represent the results of all compared methods.

It has 128 bands in the spectral range from 363 to 1018 nm, and four noisy bands were removed for experiments.

3) *Merced Dataset:* The Merced comprises HS data with spatial size 2100*180 and the connected PAN image with spatial size 6300*540, which are collected by two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

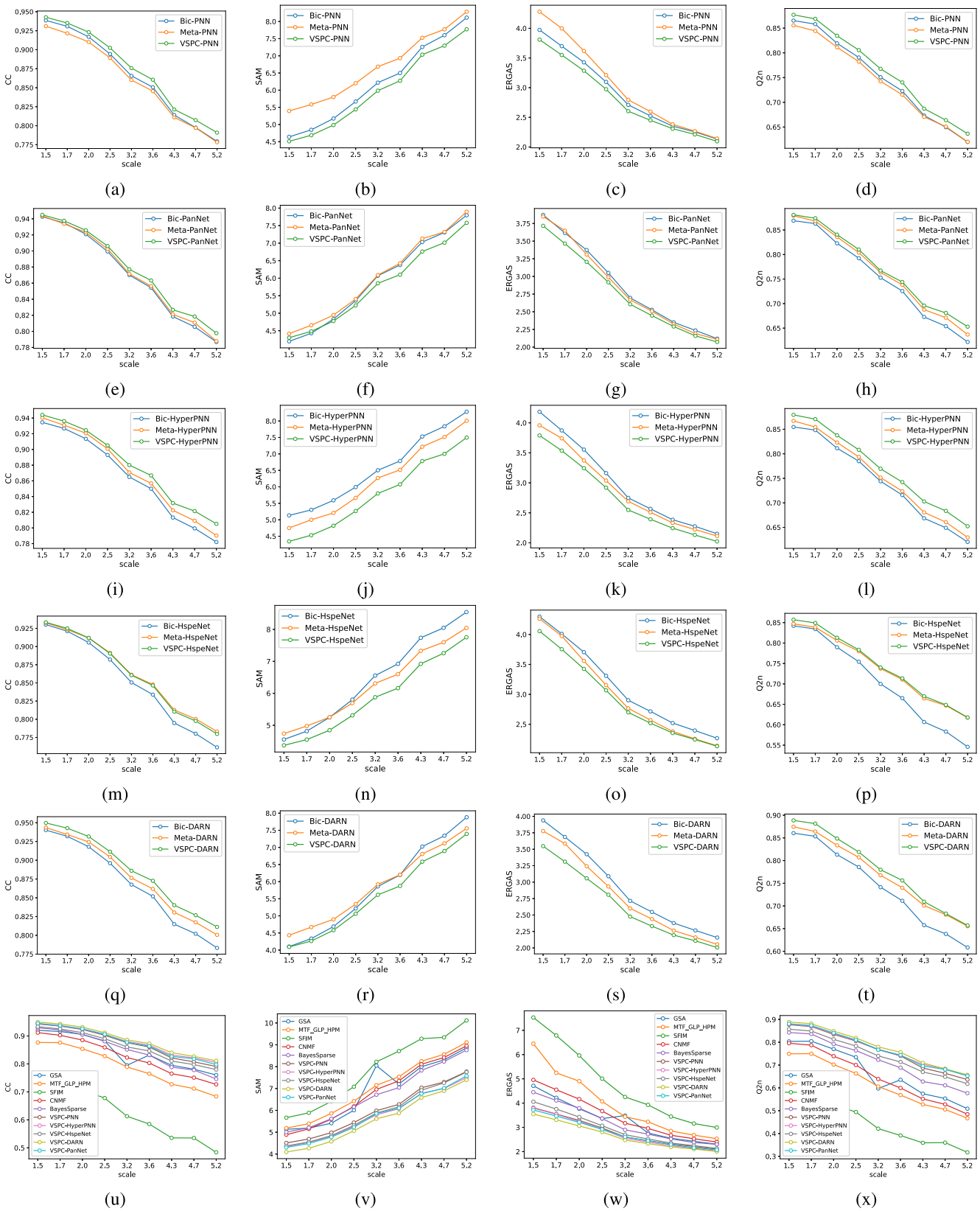


Fig. 10. Pansharpening performances under arbitrary scale s_a on the Merced dataset: (a)–(d), (e)–(h), (i)–(l), (m)–(p), and (q)–(t) are the comparative results of five specific VSPC-CNNs, respectively; (u)–(x) represent the results of all compared methods.

instruments carried on the Earth Observing One (EO-1) satellite, i.e., the Hyperion spectrometer and the Advanced Land Imager (ALI). It covers agricultural and vegetational areas of Merced, California. Specifically, the HS image consists of 242 bands, spanning from 400 to 2500 nm with a spatial resolution of 30 m,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

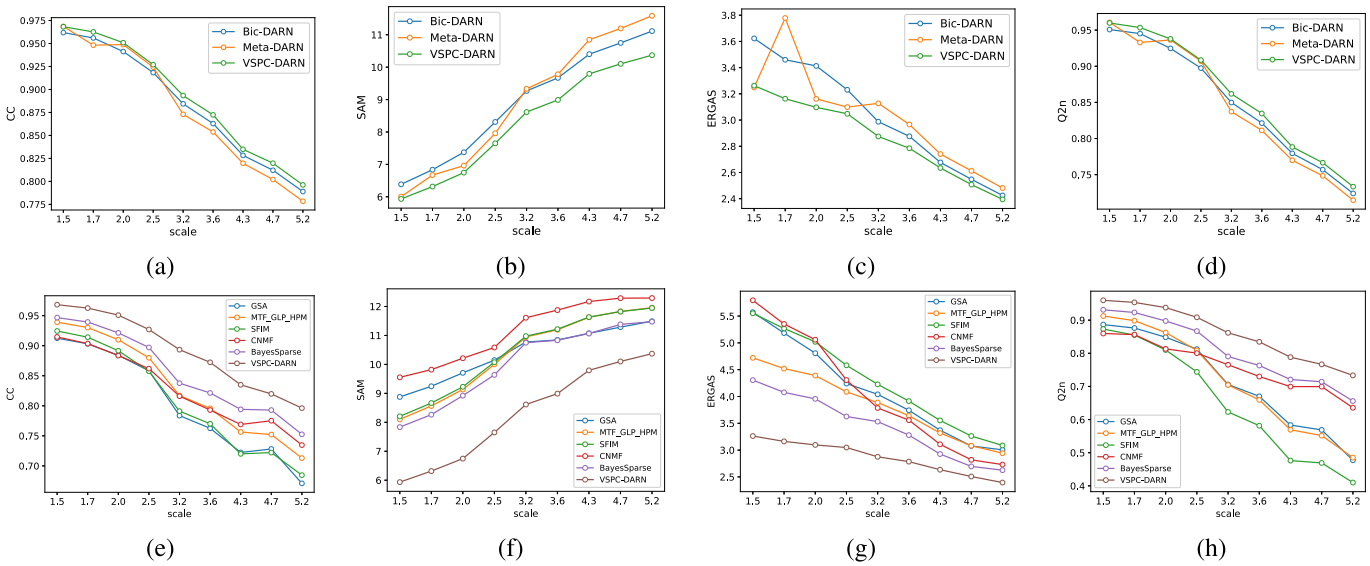


Fig. 11. Pansharpening performances under arbitrary scale s_a on the Pavia Center dataset when $s = 3$: (a)–(d) are the comparative results of VSPC-DARN; (e)–(h) represent the results of all compared methods.

and the spatial resolution of the PAN image is 10 m. After recalibration and noisy bands removal, 134 bands were left for experiments.

B. Compared Methods and Quantitative Metrics

Our VSPC-CNN includes a prepansharpening module, which could be implemented by standard CNN-based pansharpening methods. Therefore, the representative CNN, i.e., PNN [20], and four state-of-the-art CNNs, i.e., PanNet [45], HyperPNN [22], HspeNet [23], and DARN [24], are used as our prepansharpening module, which are called VSPC-PNN, VSPC-PanNet, VSPC-HyperPNN, VSPC-HspeNet, and VSPC-DARN, respectively. For more convincing comparison, five representative classical HS pansharpening methods are also considered in our experiments, including GSA [13], SFIM [14], CNMF [19], MTF-GLP-HPM [16], and Bayes sparse [18]. Since these methods were also designed for standard HS pansharpening, we attach a bicubic interpolation as the last layer of these methods to implement ARHS pansharpening as baselines, which are called Bic-PNN, Bic-PanNet, Bic-HyperPNN, Bic-HspeNet, and Bic-DARN, respectively. Besides, for CNN-based methods, we also use the metaupscale module [46] to replace the AU block in our ASAU module, which yields additional compared methods called Meta-PNN, Meta-PanNet, Meta-HyperPNN, Meta-HspeNet, and Meta-DARN, respectively. The metaupscale module was initially proposed to solve natural image super-resolution with arbitrary scale factor by dynamic convolution [46], which first extracts coordinate offsets in the low-resolution space and then uses the offsets to predict the convolution weights by dense layers. By contrast, our proposed AU block integrates VSPC to create subpixel images under required scales and then perform convolution on the subpixel image to fulfill AU.

The quality of a pansharpened HS image can be evaluated at the original observed HS scale according to RR assessment or at the PAN scale according to FR assessment [47]. RR assessment calculates the quality index between the original HS image and the pansharpened HS image, which is obtained by fusing the spatially degraded HS and PAN based on Wald's protocol [48], while FR assessment directly performs pansharpening on the original observed HS and PAN, and the quality is evaluated at PAN's scale based on the quality with no reference (QNR) protocol [49]. In order to fully evaluate our proposed method, we apply RR assessment for simulated HS datasets and both RR and FR assessment for real HS datasets. In this article, four of the most widely used RR quality metrics, namely, cross correlation (CC), spectral angle mapper (SAM), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [50], and Q2n [51], are utilized for pansharpening performance evaluation. Besides, the QNR [49], integrating

TABLE II
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE PAVIA CENTER DATASET

Methods	CC(↑)	SAM(↓)	ERGAS(↓)	Q2n(↑)
GSA	0.9220	10.2711	4.8574	0.9045
MTF-GLP-HPM	0.9511	8.1524	3.7929	0.9349
SFIM	0.9377	8.2170	4.5044	0.9026
CNMF	0.9197	9.5359	4.8515	0.9008
Bayes Sparse	0.9569	7.8864	3.4459	0.9486
Bic-PNN	0.9603	7.3253	3.3152	0.9525
Meta-PNN	0.9679	6.7601	2.9427	0.9614
VSPC-PNN	0.9664	6.7569	3.0423	0.9603
Bic-PanNet	0.9580	7.9039	3.4634	0.9481
Meta-PanNet	0.9673	6.5989	3.0064	0.9590
VSPC-PanNet	0.9680	6.4797	2.9714	0.9602
Bic-HyperPNN	0.9572	7.5323	3.4697	0.9499
Meta-HyperPNN	0.9684	6.9078	2.9425	0.9622
VSPC-HyperPNN	0.9699	6.2953	2.8500	0.9642
Bic-HspeNet	0.9582	6.6637	3.4197	0.9503
Meta-HspeNet	0.9704	6.1658	2.8544	0.9637
VSPC-HspeNet	0.9714	5.9354	2.7797	0.9651
Bic-DARN	0.9683	6.2545	2.9439	0.9622
Meta-DARN	0.9725	6.0861	2.7476	0.9661
VSPC-DARN	**0.9734**	**5.8845**	**2.6751**	**0.9678**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE CHIKUSEI DATASET

Methods	CC(↑)	SAM(↓)	ERGAS(↓)	Q2n(↑)
GSA	0.9287	4.8449	4.2181	0.8351
MTF-GLP-HPM	0.9549	3.6290	4.0318	0.8898
SFIM	0.9427	3.6705	4.9292	0.8021
CNMF	0.9276	4.3377	4.1643	0.8333
Bayes Sparse	0.9675	3.8721	3.6800	0.8553
Bic-PNN	0.9555	4.0665	3.4117	0.8740
Meta-PNN	0.9632	4.3085	3.3588	0.8933
VSPC-PNN	0.9590	4.2882	3.5897	0.8819
Bic-PanNet	0.9581	3.8942	3.1493	0.8888
Meta-PanNet	0.9623	4.2141	3.3376	0.8872
VSPC-PanNet	0.9602	4.1951	3.3742	0.8954
Bic-HyperPNN	0.9608	4.3937	3.2926	0.9000
Meta-HyperPNN	0.9629	4.5160	3.5007	0.9017
VSPC-HyperPNN	0.9672	3.7684	3.0780	0.9117
Bic-HspeNet	0.9519	**3.3278**	3.3918	0.8635
Meta-HspeNet	0.9665	3.6845	3.1419	0.9049
VSPC-HspeNet	0.9715	3.4179	2.9450	0.9175
Bic-DARN	0.9577	3.3916	3.1158	0.8899
Meta-DARN	**0.9728**	3.6487	**2.8661**	**0.9304**
VSPC-DARN	0.9693	3.5711	3.0091	0.9222

TABLE IV
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE MERCED DATASET

Methods	CC(↑)	SAM(↓)	ERGAS(↓)	Q2n(↑)
GSA	0.9565	3.7566	5.2407	0.8861
MTF-GLP-HPM	0.9054	4.2595	9.2000	0.8018
SFIM	0.9341	3.9719	6.9718	0.8438
CNMF	0.7610	4.7555	10.5327	0.6148
Bayes Sparse	0.9534	4.1744	5.6005	0.8904
Bic-PNN	0.9616	3.9033	4.8172	0.9144
Meta-PNN	0.9527	4.8796	5.4633	0.9022
VSPC-PNN	0.9630	3.9066	4.6271	0.9206
Bic-PanNet	0.9660	3.2964	4.5641	0.9172
Meta-PanNet	0.9644	3.7618	4.7803	0.9199
VSPC-PanNet	0.9651	3.7131	4.5600	0.9222
Bic-HyperPNN	0.9572	4.4794	5.2092	0.9044
Meta-HyperPNN	0.9621	4.1164	4.9048	0.9150
VSPC-HyperPNN	0.9647	3.7076	4.6038	0.9250
Bic-HspeNet	0.9562	3.6952	5.2216	0.8991
Meta-HspeNet	0.9527	4.0668	5.5283	0.8933
VSPC-HspeNet	0.9560	3.7283	5.0823	0.9064
Bic-DARN	0.9638	**3.2671**	4.6641	0.9138
Meta-DARN	0.9653	3.7950	4.5505	0.9222
VSPC-DARN	**0.9691**	3.5046	**4.2662**	**0.9329**

the spatial distortion index D_s and the spectral distortion index D_λ, is used to perform FR assessment.

C. Implementation Details

In this section, the generation of experimental data for ARHS pansharpening and key parameter settings of our proposed VSPC-CNN is presented in detail.

For each dataset, we first split it into three parts: the training region, the validation region, and the testing region. According to the training scales, we first perform the degradation process on the training and validation regions, and then partition them into patches for training, respectively. It is noteworthy that the pansharpening performance on the validation region is used to guide the setting of hyperparameters, such as the learning rate. As regards the testing region, the RR assessment and the FR assessment are applied to evaluate the performance of compared methods.

For simulated HS datasets, i.e., Pavia Center and Chikusei, the observed HS image is used as reference image $\mathbf{H}_t$. For each scale s_{ai} in training scales S, the standard HS pansharpening reference $\mathbf{H}_t^{\downarrow}(s_i')$ is yielded by Gaussian blurring and subsampling with intermediate scale s_i'. Then, the LRHS image $\mathbf{H}_t^{\downarrow}(s * s_i')$ is generated by the same spatial degradation with scale $s = 5$, and the corresponding PAN image is produced by averaging the visible bands of $\mathbf{H}_t^{\downarrow}(s_i')$. Note that $s_{ai} = s * s_i'$. Besides, we preinterpolate the LRHS image to the same spatial size as PAN via bicubic interpolation. For real HS datasets, i.e., Merced, the real observed PAN image is available, and hence, the PAN for standard HS pansharpening is yielded by the aforementioned spatial degradation with scale s_i' and $s = 3$. As the kernel size of VSPC is 3, we select a set of s_i' (from 1.0 to 2.5 with stride 0.1), and the number of training samples for each scale is 256. During training, four LR patches with size 20*20 are extracted from a randomly picked scale in training scales as a batch. Besides, we augment the patches by randomly flipping and rotating when necessary.

We implement the proposed method with the Pytorch framework and train them on a PC with an NVIDIA GeForce GTX 2060 super GPU. In the training process, the AdamW [52] optimizer with weight decay 10^{-4} is employed, and the optimization is terminated at the 300th epoch. The learning rate is initially set to 6×10^{-4}. Furthermore, traditional pansharpening methods, including GSA, MTF-GLP-HPM, SFIM, CNMF, and BayesSparse, are implemented according to the code provided by the following publication [7]. For CNN-based methods, the network parameters' settings of networks used in the prepansharpening modules strictly follow the related publications. The filter number of VSPC in our ASAU module is set to 64, and the kernel size is 3. In order to make a relatively fair comparison, the same optimizer setting for all those CNN-based methods, i.e., an AdamW optimizer with weight decay 10^{-4}, is adopted. Besides, the initial learning rates of those methods are all set to 6×10^{-4}. As regards training sample setting, the spatial size of each patch is set to 20×20, and the total number of the training samples is 4096 since the number of training samples for each scale in the range of [1.0, 2.5] with interval 0.1 is 256.

D. Arbitrary-Resolution HS Pansharpening

In this section, we focus on the presentation of arbitrary-resolution HS pansharpening experimental results at RR. Specifically, we randomly select $s' = 1.5, 1.7, 2.0, 2.5, 3.2, 3.6, 4.3, 4.7, 5.2$ as test scales for quantitative assessment on three datasets. Note that, in RR experiments, we use the observed HS image as a reference and degrade it with selected scales, which would generate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

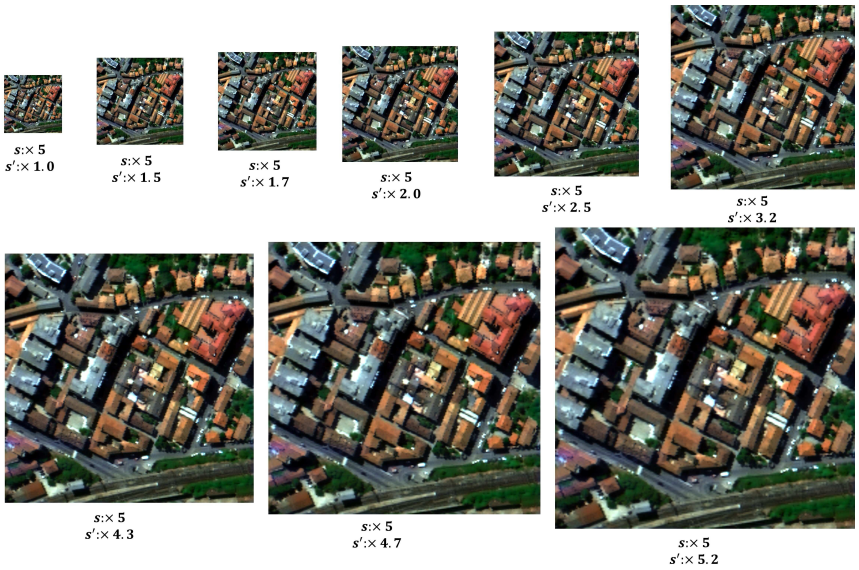


Fig. 12. Visual comparison of the PaviaC dataset, which is pansharpened with different pansharpening scales by our proposed method.

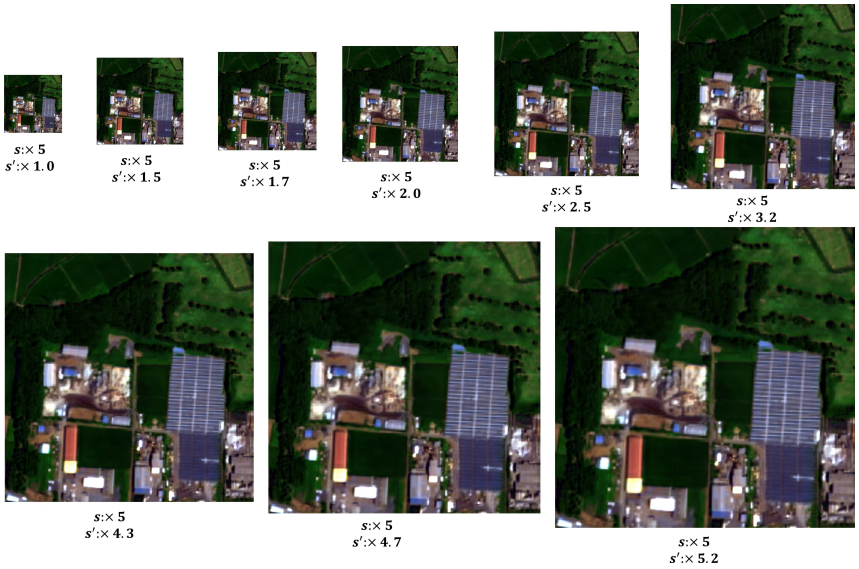


Fig. 13. Visual comparison of the Chikusei dataset, which is pansharpened with different pansharpening scales by our proposed method.

different LRHS images as inputs to CNN. The outputs of all compared methods for different scales are, finally, evaluated by the aforementioned quality metrics. The experimental results on the Pavia Center, Chikusei, and Merced datasets are illustrated in Figs. 8–10, respectively.

As can be seen, for bicubic-attached CNNs, since bicubic interpolation could not introduce useful textures or details to desired HS images, the pansharpening performances are limited. Our proposed VSPC-CNNs win better CC, SAM, ERGAS, and Q2n indexes than the corresponding Bic-CNNs and Meta-CNNs, especially the SAM. It means that our proposed method can better keep the spectral fidelity of pansharpened HS images. It is obvious that the CNN-based methods outperform the classical pansharpening ones. Besides, it can also be observed that, as the test scale increases, quality metrics, such as CC, SAM, and Q2n, tend to get worse, which implies that the greater the difference in spatial resolution, the harder for CNNs to reconstruct ARHS images. It is noteworthy that ERGAS has the opposite behavior as the changes of root mean square error (RMSE) are small with the increase of scale, which are divided by RMSE in the formula of ERGAS. In addition, notice that we set prepansharpening scale $s = 5$ for the simulated datasets and $s = 3$ for the real dataset. The scale for the real dataset depends on the ratio of the spatial resolutions between the PAN image and the HS image. We set the scale $s = 5$ for the simulated datasets without special considerations, relatively randomly, since they do not have real PANs. For a more comprehensive evaluation, we additionally conduct the experiments with the scale $s = 3$ on the Pavia Center dataset, and the experimental results are graphically illustrated in Fig. 11. As shown in this figure, our VSPC-DARN also achieves excellent performance under $s = 3$, which is similar to the case of the scale $s = 5$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

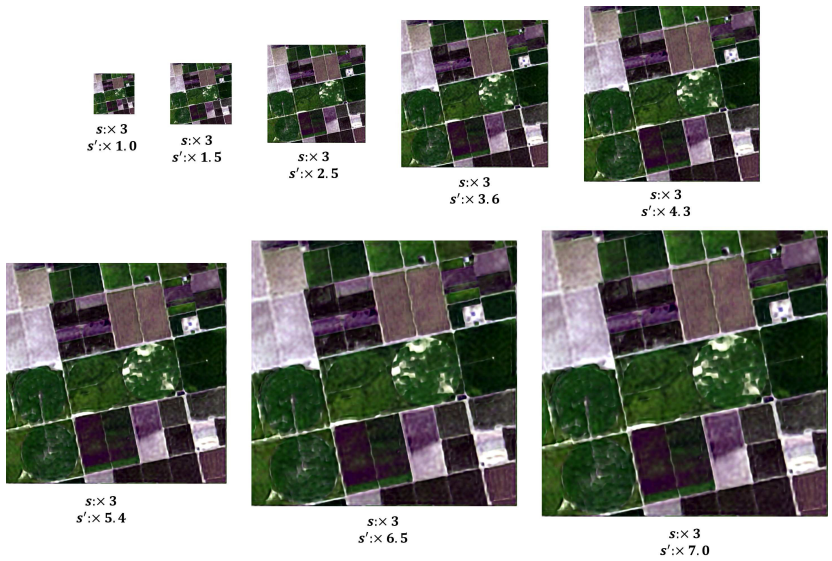


Fig. 14. Visual comparison of the Merced dataset, which is pansharpened with different pansharpening scales by our proposed method.



Fig. 15. Visual comparison among pansharpening results from different tested methods of the Merced dataset. The input is a 100*100 LRHS and a 300*300 PAN, while red boxes in the LRHS and the PAN indicate the crop area for demonstration. (a) GSA + Bic. (b) SFIM + Bic. (c) CNMF + Bic. (d) Bayes sparse + Bic. (e) Bic-PNN. (f) VSPC-DARN.

Moreover, for visual comparison, the ARHS pansharpening results of the VSPC-DARN are visualized in Figs. 12–14, as a representation of our proposed VSPC-CNN, where ARHS pansharpening scales within and beyond the training scale range are involved. In the pansharpening phase, the trained VSPC-DARN takes the LRHS and PAN images as inputs and yields HS images with different resolutions according to the user-customized scales. Here, the user-customized scales ranging from 5.0 to 26.0 for the Pavia Center and Chikusei datasets (3.0–21.0 for the Merced) are chosen for visualization. It is notorious that the reconstruction difficulty of the target resolution HS image is growing as the scale increases. In general, most existing pansharpening methods handle the scales ranging from 3 to 8. However, our proposed VSPC-CNN is able to generate HS images with arbitrary resolution under a single trained model, including those resolutions much higher than the LRHS, e.g., $s_a = 26.0$.

E. Standard HS Pansharpening

Standard HS pansharpening is a special case of ARHS pansharpening when the target pansharpening resolution is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

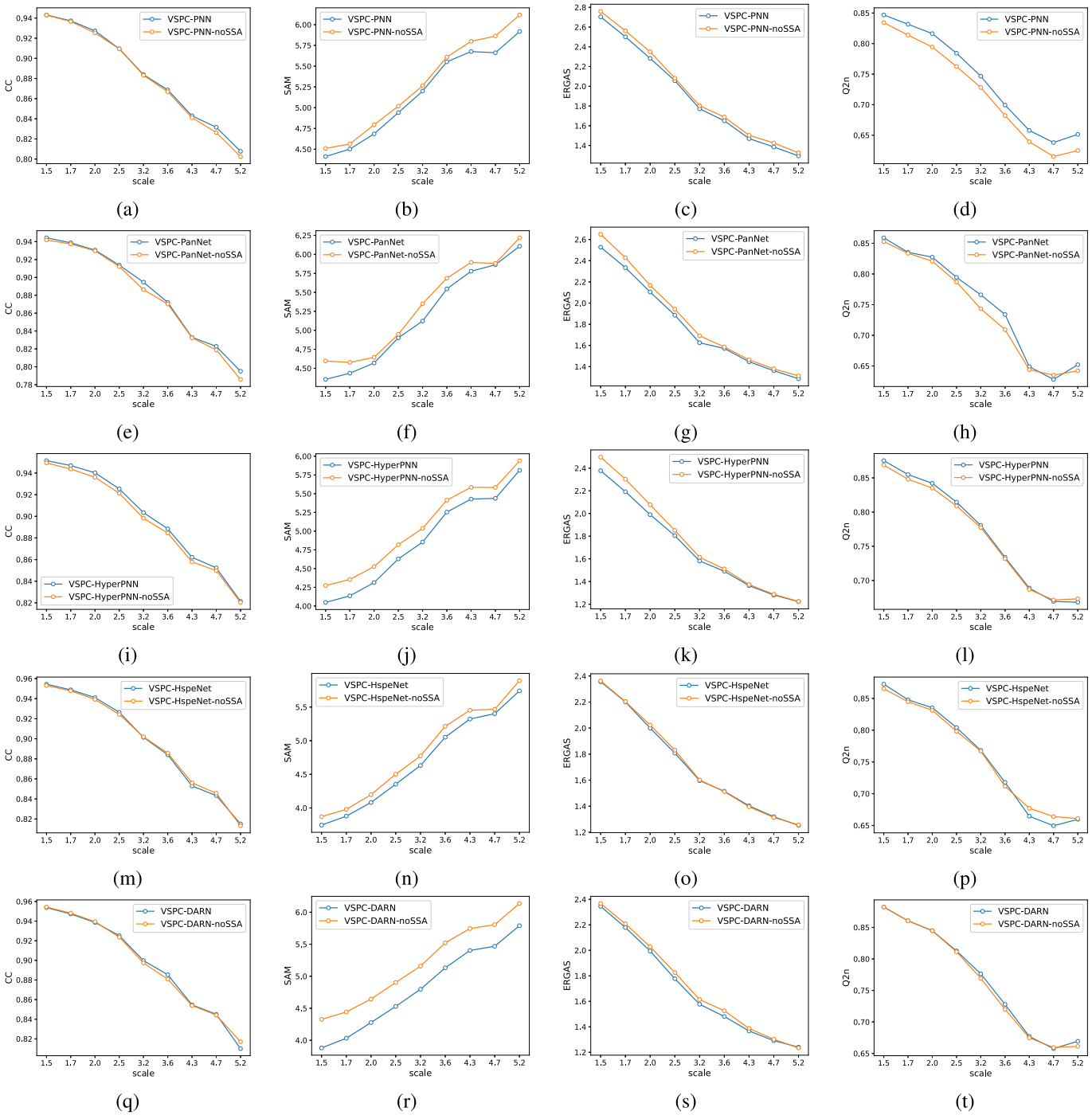


Fig. 16. Ablation study of the proposed SSA block on the Chikusei dataset: (a)–(d), (e)–(h), (i)–(l), (m)–(p), and (q)–(t) are the comparative results of five specific VSPC-CNNs with and without SSA block, respectively.

equal to that of the PAN, i.e., the intermediate scale s' is set to 1. The quantitative results of different competing methods on Pavia Center, Chikusei, and Merced datasets are listed in Table II–IV, with the best value for each quality metric highlighted in boldface and the second-best value underlined.

As can be seen in the tables, Bayes sparse achieves the best results on Pavia Center and Chikusei datasets while a little behind on the Merced dataset among the traditional pansharpening methods. It can be apparently observed that CNN-based methods are superior to traditional competing ones. For standard pansharpening, our proposed VSPC-CNNs produce comparable or even better CC, SAM, ERGAS, and Q2n values than their corresponding Bic-CNNs and Meta-CNNs, which implies that our proposed method for ARHS pansharpening can also deal with standard pansharpening well. In addition, the performance of CNN-based methods will get better as the prepansharpening module becomes more complicated.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

TABLE V
QUALITY METRICS OF FR EXPERIMENT ON THE MERCED DATASET

Methods	D_λ	D_s	QNR
GSA	0.1112	0.0987	0.8011
MTF-GLP-HPM	0.1052	0.1243	0.7835
SFIM	0.0623	0.0809	0.8618
CNMF	0.1109	0.1949	0.7158
Bic-PNN	0.0254	0.0550	0.9210
Meta-PNN	0.0474	0.0429	0.9118
VSPC-PNN	0.0337	0.0476	0.9203
VSPC-PNN-FT	0.0315	0.0475	0.9224
Bic-PanNet	0.0275	**0.0381**	0.9354
Meta-PanNet	0.0335	0.0435	0.9244
VSPC-PanNet	0.0341	0.0438	0.9236
VSPC-PanNet-FT	0.0344	0.0438	0.9233
Bic-Hyper-PNN	0.0591	<u>0.0414</u>	0.9019
Meta-HyperPNN	0.0356	0.0425	0.9234
VSPC-HyperPNN	0.0234	0.0443	0.9333
VSPC-HyperPNN-FT	0.0240	0.0432	0.9339
Bic-HspeNet	0.0164	0.0567	0.9279
Meta-HspeNet	0.0424	0.0418	0.9176
VSPC-HspeNet	0.0284	0.0600	0.9132
VSPC-HspeNet-FT	0.0275	0.0600	0.9142
Bic-DARN	**0.0082**	0.0494	**0.9428**
Meta-DARN	0.0230	0.0430	0.9350
VSPC-DARN	0.0146	0.0451	0.9410
VSPC-DARN-FT	<u>0.0145</u>	0.0450	<u>0.9411</u>

F. Full-Resolution Experiment on Real Hyperspectral Dataset

To evaluate the generalization capability of the proposed method, FR experiments are also conducted on the real dataset, i.e., the Merced dataset. Note that the Bayes sparse used for comparison in RR experiments is omitted as the blur matrix (necessary for implementation) is unavailable.

Table V lists the quantitative results of the no-reference metrics D_λ, D_s, and QNR for different competing methods, as described in Section III-B. By comparison, the experimental results show similar results to those of standard HS pansharpening in RR experiments. It is noteworthy that, for a more comprehensive comparison, the target adaptive technique is also included in our experiments, where it serves as prepansharpening in our VSPC-CNN scheme as PNN, PanNet, HyerPNN, HspeNet, and DARN do. The target adaptive was proposed by Scarpa *et al.* [53] mainly to solve the mismatch between training data and target data. We apply the target adaptive strategy on the Merced dataset, the results of which are included in Table V. Notice that FT means fine-tuning for the target adaptive. As can be observed, the performance gain of the target adaptive is limited. This may be due to that our training data and target data are from the same sensor and the same scene, which was also pointed out in [53].

Furthermore, we graphically illustrate the ARHS pansharpening results from different competed methods, including GSA, SFIM, CNMF, Bayes sparse, Bic-PNN, and the proposed VSPC-DARN for qualitative comparison, as shown in Fig. 15. Notice that competed methods are attached with a bicubic interpolation as the terminal step to accomplish the final ARHS pansharpening results. In this experiment, we select $s_a = 5.4$ (i.e., $s = 3$ and $s' = 1.8$), $s_a = 7.5$ (i.e., $s = 3$ and $s' = 2.5$), $s_a = 10.5$ (i.e., $s = 3$ and $s' = 3.5$), $s_a = 13.5$ (i.e., $s = 3$ and $s' = 4.5$), and $s_a = 19.5$ (i.e., $s = 3$ and $s' = 6.5$) as the target pansharpening scales for visualization. It can be observed that our VSPC-DARN (as a representative instance of our proposed VSPC-CNN) has a better performance for the structure part, especially when the pansharpening scale becomes larger.

G. Ablation Study

To gain better convincingness and understanding of the proposed SSA block, we have conducted some ablation studies on the Chikusei dataset. From the experiment results in Fig. 16, VSPC-CNNs without SSA block suffer performance degradation (especially the SAM) compared with those including SSA block. Since different scales are related to different downsampling degradations in ARHS pansharpening, introducing the scale factor into CNN learning could have a positive effect on ARHS pansharpening. Therefore, both intuitive analysis and experiment results validate the effectiveness of the proposed SSA block integrated into the ASAU module.

IV. CONCLUSION

For the ARHS pansharpening task that aims to pansharpen HS images to any user-customized scales, one of the main challenges is how to generate HS images beyond the training scales using merely a single pansharpening model. In this work, we design an AU-CNN framework composed of a two-stage elevating thread and accordingly develop a specific VSPC-CNN to address the challenge of the ARHS pansharpening. Our innovative VSPC-based ARHS pansharpening CNN comprises two major module: 1) a prepansharpening module that is used to improve the spatial resolution of the HS image to that of the PAN and 2) a VSPC-based ASAU module that encapsulates the VSPC unit and the SSA block for further resolution elevating. It is able to generalize a CNN model trained under certain scales to pansharpen HS data to any scales. Comprehensive experiments on both simulated and real HS datasets demonstrate the superiority of our proposed VSPC-CNN for ARHS pansharpening in terms of numeric and visual comparisons. In the future, some techniques, such as target adaptive that could enhance predictive robustness, could be taken into consideration in our ARHS method design to help deal with scale generalization.

REFERENCES

[1] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.
[2] Z. Wu *et al.*, "Scheduling-guided automatic processing of massive hyperspectral image classification on cloud computing architectures," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3588–3601, Jul. 2021.
[3] Z. Wu, J. Sun, Y. Zhang, Z. Wei, and J. Chanussot, "Recent developments in parallel and distributed computing for remotely sensed big data processing," *Proc. IEEE*, vol. 109, no. 8, pp. 1282–1305, Aug. 2021.
[4] P. Du, S. Liu, J. Xia, and Y. Zhao, "Information fusion techniques for change detection from multi-temporal remote sensing images," *Inf. Fusion*, vol. 14, no. 1, pp. 19–27, Jan. 2013.
[5] A. Mohammadzadeh, A. Tavakoli, and M. J. V. Zoej, "Road extraction based on fuzzy logic and mathematical morphology from pan-sharpened IKONOS images," *Photogramm. Rec.*, vol. 21, no. 113, pp. 44–60, Mar. 2006.
[6] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

[7] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[8] G. Vivone *et al.*, "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2014.

[9] G. Vivone *et al.*, "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[10] L. He *et al.*, "Pansharpening via detail injection based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1188–1204, Apr. 2019.

[11] G. Vivone, R. Restaino, G. Licciardi, M. D. Mura, and J. Chanussot, "Multiresolution analysis and component substitution techniques for hyperspectral pansharpening," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 2649–2652.

[12] P. Chavez, S. C. Sides, and J. A. Anderson, "Comparison of three different methods to merge multiresolution and multispectral data-Landsat TM and SPOT panchromatic," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 3, pp. 295–303, Mar. 1991.

[13] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of ms + pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Sep. 2007.

[14] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Nov. 2010.

[15] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[16] G. Vivone, R. Restaino, M. D. Mura, G. Licciardi, and J. Chanussot, "Contrast and error-based fusion schemes for multispectral image pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 930–934, May 2013.

[17] Q. Wei, N. Dobigeon, and J. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[18] Q. Wei, N. Dobigeon, and J. Y. Tourneret, "Bayesian fusion of multi-band images," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1117–1127, Sep. 2015.

[19] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2011.

[20] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[21] Y. Wei, Q. Yuan, H. Shen, and L. Zhang, "Boosting the accuracy of multispectral image pansharpening by learning a deep residual network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1795–1799, Oct. 2017.

[22] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[23] L. He, J. Zhu, J. Li, D. Meng, J. Chanussot, and A. Plaza, "Spectral-fidelity convolutional neural networks for hyperspectral pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5898–5914, 2020.

[24] Y. Zheng, J. Li, Y. Li, G. Jie, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

[25] W. Dong, T. Zhang, J. Qu, S. Xiao, J. Liang, and Y. Li, "Laplacian pyramid dense network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[26] J. Qu, Y. Shi, W. Xie, Y. Li, X. Wu, and Q. Du, "MSSL: Hyperspectral and panchromatic images fusion via multiresolution spatial–spectral feature learning networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[27] P. Wang, L. Wang, Y. Wu, and H. Leung, "Utilizing pansharpening technique to produce sub-pixel resolution thematic map from coarse remote sensing image," *Remote Sens.*, vol. 10, no. 6, p. 884, Jun. 2018.

[28] F. Laporterie-Déjean, H. de Boissezon, G. Flouzat, and M.-J. Lefèvre-Fonollosa, "Thematic and statistical evaluations of five panchromatic/multispectral fusion methods on simulated PLEIADES-HR images," *Inf. Fusion*, vol. 6, no. 3, pp. 193–212, Sep. 2005.

[29] T. Ranchin and L. Wald, "Benefits of fusion of high spatial and spectral resolutions images for urban mapping," in *Proc. 26th Int. Symp. Remote Sens. Environ. 18th Annu. Symp. Can. Remote Sens. Soc.*, 1996, pp. 262–265.

[30] A. Jenerowicz and M. Woroszkiewicz, "The pan-sharpening of satellite and UAV imagery for agricultural applications," *Proc. SPIE*, vol. 9998, Oct. 2016, Art. no. 99981S.

[31] E. Ibarrola-Ulzurrun, C. Gonzalo-Martín, and J. Marcello, "Influence of pansharpening in obtaining accurate vegetation maps," *Can. J. Remote Sens.*, vol. 43, no. 6, pp. 528–544, 2017.

[32] J. K. Gilbertson, J. Kemp, and A. Van Niekerk, "Effect of pan-sharpening multi-temporal Landsat 8 imagery for crop type differentiation using different classification techniques," *Comput. Electron. Agricult.*, vol. 134, pp. 151–159, Mar. 2017.

[33] Y. Qu, H. Qi, B. Ayhan, C. Kwan, and R. Kidd, "DOES multispectral/hyperspectral pansharpening improve the performance of anomaly detection?" in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 6130–6133.

[34] P. S. Huang and T.-M. Tu, "A target fusion-based approach for classifying high spatial resolution imagery," in *Proc. IEEE Workshop Adv. Techn. Anal. Remotely Sensed Data*, Oct. 2003, pp. 175–181.

[35] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Jun. 2019.

[36] M. D. Zeiler, D. Krishnan, G. W. Taylor, and R. Fergus, "Deconvolutional networks," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2528–2535.

[37] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2014, pp. 818–833.

[38] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1874–1883.

[39] W. Shi *et al.*, "Is the deconvolution layer the same as a convolutional layer?" 2016, *arXiv:1609.07009*.

[40] N. Ahn, B. Kang, and K.-A. Sohn, "Fast, accurate, and lightweight super-resolution with cascading residual network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 252–268.

[41] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[42] A. Mehri, P. B. Ardakani, and A. D. Sappa, "MPRNet: Multi-path residual network for lightweight image super resolution," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 2704–2713.

[43] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

[45] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 5449–5457.

[46] X. Hu, H. Mu, X. Zhang, Z. Wang, T. Tan, and J. Sun, "Meta-SR: A magnification-arbitrary network for super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 1575–1584.

[47] G. Vivone, M. D. Mura, A. Garzelli, and F. Pacifici, "A benchmarking protocol for pansharpening: Dataset, preprocessing, and quality assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6102–6118, 2021.

[48] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[49] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[50] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?" in *Proc. 3rd Conf. Fusion Earth Data, Merging Point Meas., Raster Maps Remotely Sensed Images*, 2000, pp. 99–103.

[51] A. Garzelli and F. Nencini, "Hypercomplex quality assessment of multi/hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 662–665, Oct. 2009.

[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.

[53] G. Scarpa, S. Vitale, and D. Cozzolino, "Target-adaptive CNN-based pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5443–5457, Sep. 2018.



Lin He (Member, IEEE) received the B.S. degree from the Xi'an Institute of Technology, Xi'an, China, in 1995, the M.S. degree from Chongqing University, Chongqing, China, in 2003, and the Ph.D. degree in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, in 2007.

Since 2007, he has been with the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, where he is currently a Full Professor. His research interests include hyperspectral image processing, unmanned aerial vehicle (UAV) image processing, spectral–spatial–temporal information fusion, machine learning, and high-dimensional signal processing.



Jinhua Xie received the B.S. degree from the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, in 2020, where he is currently pursuing the M.S. degree.

His research interests include degraded information reconstruction from hyperspectral images, data fusion, and machine learning in remote sensing.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

From 2013 to 2021, she was a Full Professor with Sun Yat-sen University. Since 2022, she has been with the China University of Geosciences as a Full Professor. She has obtained several prestigious funding grants at the national and international levels. She has published more than 160 journal citation report (JCR) papers, 60 international conference papers, and a book chapter.

Dr. Li has been serving as the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING since 2021.



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is currently a Full Professor and the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura. He has authored more than 600 publications, including 284 journal citation report (JCR) journal papers (more than 170 in IEEE journals), 24 book chapters, and over 300 peer-reviewed conference proceeding papers. His main research interests comprise hyperspectral data processing and parallel computing of remote sensing data.

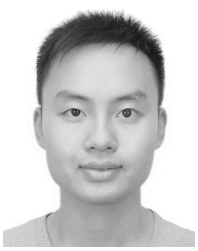
Dr. Plaza was also a member of the Steering Committee of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He was a member of the Editorial Board of the IEEE Geoscience and Remote Sensing Newsletter from 2011 to 2012 and the *IEEE Geoscience and Remote Sensing Magazine* in 2013. He is also a fellow of the IEEE "for contributions to hyperspectral data processing and parallel computing of Earth observation data." He was a recipient of the Best Column Award of the *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of the JSTARS journal, and the Most Highly Cited Paper (2005–2010) of the *Journal of Parallel and Distributed Computing*. He received best paper awards at the IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He was a recipient of the recognition of Best Reviewer of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he has served as Associate Editor from 2007 to 2012. He was included in the 2018 and 2019 Highly Cited Researchers List (Clarivate Analytics). He has guest-edited ten special issues on hyperspectral remote sensing for different journals. He is also an Associate Editor for IEEE ACCESS. He has served as the Director of Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President of the Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for over 50 different journals. He has served as the Editor-in-Chief of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING journal from 2013 to 2017. Additional information: http://www.umbc.edu/rssipl/people/aplaza.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA, the KTH Royal Institute of Technology, Stockholm, Sweden, and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008. He was a member of the Institut Universitaire de France from 2012 to 2017. He has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018. He received multiple outstanding paper awards. He was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRS-S Chapter Excellence Award. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). He was the Chair and the Co-Chair of the GRS Data Fusion Technical Committee from 2009 to 2011 and 2005 to 2008, respectively. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE.



Jiawei Zhu received the B.E. and M.S. degrees from the School of Automation Science and Engineering, South China University of Technology, Guangzhou, China, in 2018 and 2021, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include brain signal processing, remote sensing image processing, data fusion, and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:58:07 UTC from IEEE Xplore. Restrictions apply.